

PAPER_1352 — Quantum Spin Hall Effect

Framework: UQFF — Star-Magic v5.27+ | **Tier:** F Condensed Matter (CLOSED) **Calculator:**
`calculate_paradox({"paradox": "quantum_spin_hall"})`

Master Expression

$\sigma_{xy_spin} = e^2/h$ per spin; protected by $D_{BSFG}-1=5$ boundary

Match: STRUCTURAL

Interpretation

Edge currents topologically protected by $D_{BSFG} - 1 = 5$ boundary dimension. Time reversal via F_{TRZ} doubling.

UQFF Locked Primitives

- $D_{phys} = 4$, $D_{BSFG} = 6$, $D_{crit} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{CH} = 9$
- $K_{MEX} = 25/12$, $\beta_i = 0.6029$, $\Phi_{res} = 0.84$, $\Lambda = 0.00729735$, $F_{TRZ} = 0.1$
- $\omega_{SCm} = 1.25$ THz

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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